



JSS
SCIENCE AND
TECHNOLOGY
UNIVERSITY
MYSURU

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead institution for World Bank Assistance under TEQIP Scheme



DEPARTMENT OF MECHANICAL ENGINEERING

I / II Semester B.E. Degree

Elements of Mechanical Engineering [22ME110/22ME210]

Faculty Name: Karthik MS

QUESTION BANK

UNIT - 3

Belt Drives:

- (1) Define belt drives, Sketch and explain: (i) Open belt drive (ii) Cross belt drive.
- (2) Define: (i) Velocity ratio of belt drive (ii) Creep (iii) Slip.

Gear Drives:

- (1) Define gear drives and terminology of gears.
- (2) Sketch and explain and list the advantages and disadvantages of: (i) spur gear (ii) helical gear (iii) Rack and pinion (iv) bevel gear (v) worm and worm wheel.
- (3) Explain with a neat sketch the arrangement of Simple and compound gear train and derive an expression for velocity ratio.

Metal Joining Process: (Soldering, Brazing and Welding)

- (1) Define soldering, brazing and welding.
- (2) List the difference between soldering, brazing and welding.
- (3) List the advantages and disadvantages of soldering, brazing and welding.
- (4) Explain with a neat sketch, electric arc welding method.
- (5) Define gas ratio and Explain with a neat sketch, oxy-acetylene gas welding method.
- (6) Discuss with a neat sketch different types of flames, which is produced in oxy-acetylene process.
- (7) List the advantages and disadvantages of oxy-acetylene gas welding.
- (8) List the advantages and disadvantages of electric arc welding.